

Mon 13

$$y'' + 2y' - 8y = 0$$

$$\lambda^2 + 2\lambda - 8 = 0$$

$$(\lambda + 1)^2 - 9 = 0$$

$$\lambda + 1 = \pm \sqrt{9}$$

$$\lambda = -4 \text{ or } 2$$

Name

- For the ODE $y'' + 2y' - 8y = t$

- Write down the complementary sol.

$$y_c(t) = C_1 e^{-4t} + C_2 e^{2t}$$

- Compute a particular sol.

$$y_p(t) = -\frac{1}{8}t - \frac{1}{32}$$

- Describe the solution as $t \rightarrow \infty$

As $t \rightarrow \infty$
 $C_2 e^{2t}$ grows exp!!

$$y_p = a_1 t + a_0$$

$$y_p' = a_1, y_p'' = 0$$

$$0 + 2(a_1) - 8(a_1 t + a_0) = t$$

$$-8a_1 = 1 \quad a_1 = -\frac{1}{8}$$

$$2a_1 - 8a_0 = 0$$

$$-\frac{1}{4} = 8a_0 \Rightarrow a_0 = -\frac{1}{32}$$

1. For the ODE $y'' + 4y' + 13y = e^t$

- 1.1. Write down the complementary sol.

$$y_c(t) =$$

- 1.2. Compute a particular sol.

$$y_p(t) =$$

- 1.3. Describe the solution as $t \rightarrow \infty$

As $t \rightarrow \infty$

$$2\lambda^2 - 4\lambda + 20 = 0$$

$$\lambda^2 - 2\lambda + 10 = 0$$

$$(\lambda - 1)^2 + 9 = 0$$

$$\lambda - 1 = \pm 3i, \lambda = 1 \pm 3i$$

- For the ODE $2y'' - 4y' + 20y = 40e^t$ with

- Write down the complementary sol.

$$y_c(t) = c_1 e^t \sin(3t) + c_2 e^t \cos(3t)$$

- Compute a particular sol.

$$y_p(t) = \frac{40}{18} e^t$$

- Describe the solution as $t \rightarrow \infty$

As $t \rightarrow \infty$ everything grows like e^t . y_1 and y_2 wiggle like $\cos(3t)$.

2. For the ODE $y'' + 2y' + 5y = 17e^{-t/2}$

- 2.1. Write down the complementary sol.

$$y_c(t) =$$

- 2.2. Compute a particular sol.

$$y_p(t) =$$

- 2.3. Describe the solution as $t \rightarrow \infty$

As $t \rightarrow \infty$

$$y_p = a e^t$$

$$y_p' = a e^t, y_p'' = a e^t$$

$$2a e^t - 4a e^t + 20a e^t = 40e^t$$

$$(2 - 4 + 20)a = 40$$

$$a = 40/18$$